Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across multiple product categories. The objective is to generate actionable insights into spending patterns, customer segmentation, product preferences, and subscription behavior to support strategic business decision-making.

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2. Dataset Summary

- **Rows:** 3,900
- **Columns:** 18

Key Features:

- **Customer Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Item Purchased, Category, Purchase Amount, Season, Size, Color
- **Shopping Behavior:** Discount Applied, Promo Code Used, Previous Purchases, Purchase Frequency, Review Rating, Shipping Type
- **Missing Data:** 37 missing values identified in the *Review Rating* column

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3. Exploratory Data Analysis (Python)

The analysis began with data preparation and cleaning using Python:

- **Data Loading:** The dataset was imported using `pandas`.
- **Initial Exploration:**
  - `df.info()` was used to inspect the dataset structure
  - `.describe()` was used to generate summary statistics

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discou Appli
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	N
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	N
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	N
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	N
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	N
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	N
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	N

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

- **Missing Data Handling:**
  Null values were identified and handled by imputing missing values in the `Review Rating` column using the median rating within each product category.

- **Column Standardization:**
  Column names were converted to *snake_case* to improve readability and maintain consistency.

- **Feature Engineering:**
  - Created an **age_group** column by grouping customer ages into bins
  - Generated a **purchase_frequency_days** variable based on purchase data

- **Data Consistency Check:**
  Verified redundancy between `discount_applied` and `promo_code_used` subsequently dropped `promo_code_used`

- **Database Integration:**
  Established a connection to PostgreSQL and loaded the cleaned dataset into the database for further SQL-based analysis.

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4. Data Analysis Using SQL (Business Transactions)

Structured queries were executed in PostgreSQL to address key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male versus female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. **High-Spending Discount Users** – Identified customers who utilized discounts while still spending above the average purchase value.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88

Total rows: 839 Query complete 00:00:(

3. **Top 5 Products by Rating** – Determined the products with the highest average review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping Type Comparison** – Analyzed differences in average purchase value between Standard and Express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. **Subscribers vs. Non-Subscribers** – Compared both average spending and total revenue based on subscription status.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. **Discount-Dependent Products** – Identified the five products with the highest proportion of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. **Customer Segmentation** – Categorized customers into *New*, *Returning*, and *Loyal* segments based on purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. **Top 3 Products per Category** – Identified the most frequently purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Evaluated whether customers with more than five purchases are more likely to subscribe.

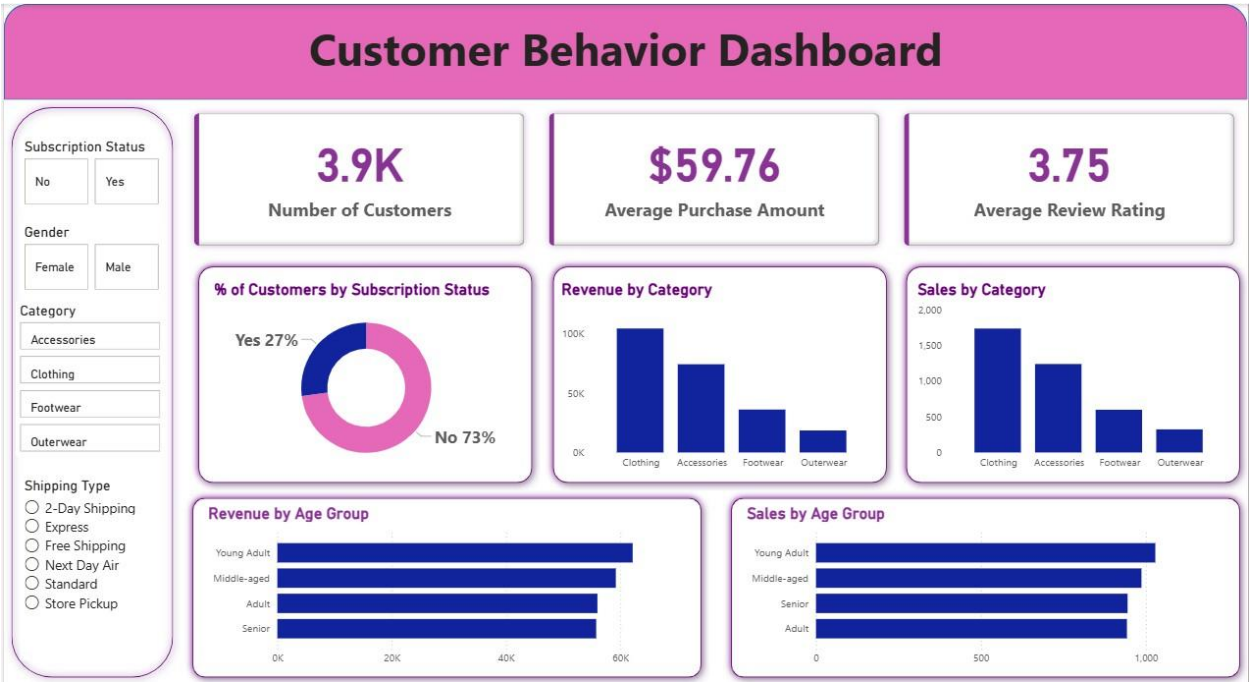
	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. **Revenue by Age Group** – Calculated total revenue contribution across different age groups

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Power BI Dashboard

An interactive dashboard was developed in Power BI to visually present key insights, enabling intuitive exploration of customer behavior and business performance metrics.



6. Business Recommendations

- **Enhance Subscription Growth:** Promote exclusive benefits to increase subscriber acquisition and retention.
- **Strengthen Customer Loyalty Programs:** Incentivize repeat purchases to transition customers into the *Loyal* segment.
- **Optimize Discount Strategy:** Reassess discount policies to balance revenue growth with profit margins.
- **Improve Product Positioning:** Highlight top-performing and highly rated products in marketing campaigns.
- **Implement Targeted Marketing:** Focus efforts on high-revenue age groups and customers who prefer express shipping.